Why if $f: M \to N$ $\quad df_a : T_a M \to T_{f(a)} N$?

try $\quad M = U \underset{\text{open}}{\subset} \mathbb{R}^m \qquad N = V \underset{\text{open}}{\subset} \mathbb{R}^n$

$T_a M \equiv \mathbb{R}^m \qquad T_{f(a)} N = \mathbb{R}^n$

$f: U \to V$ differentiable at $a \in U$ if:

$\exists$ linear map $df_a : \mathbb{R}^m \to \mathbb{R}^n$ s.t.

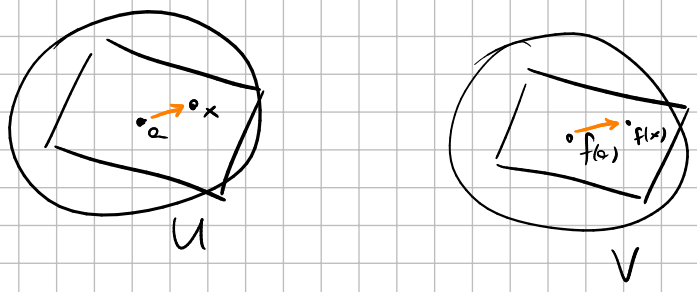
$$\lim_{x \to a} \frac{\| f(x) - f(a) - df_a(x-a) \|}{\| x - a \|} = 0$$

vector at $f(a)$

vector at $f(a)$

if f is C^1 then $df_a(x-a) = \nabla f(a) \cdot (x-a)$

$df_a : \mathbb{R}^n \to \mathbb{R}^n$



group automorphism

$$\mathcal{C} : G \to \mathrm{Aut}(G) \qquad\qquad \mathcal{C}_g : G \to G$$
$$g \mapsto \mathcal{C}_g \qquad\qquad h \mapsto g h g^{-1}$$

$$\mathrm{Ad} : G \to \mathrm{Aut}(\mathfrak{g}) \qquad\qquad \mathrm{Ad}_g : \mathfrak{g} \to \mathfrak{g}$$
$$g \mapsto \mathrm{Ad}_g \qquad\qquad A \mapsto \frac{d}{dt}\left(g e^{tA} g^{-1}\right)\Big|_{t=0}$$
$$= \frac{d}{dt} \mathcal{C}_g(e^{tA})\Big|_{t=0}$$

adjoint representation of G

if we are working with matrices $\mathrm{Ad}_g(A) = g A g^{-1}$

$\mathrm{ad} : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ associated Lie algebra rep.

$$\mathrm{ad}_A(B) = \frac{d}{dt}\left(\mathrm{Ad}_{e^{tA}}(B)\right)\Big|_{t=0} = \frac{d}{dt}\left(e^{tA} B e^{-tA}\right)\Big|_{t=0}$$
$$= AB - BA = [A, B]$$

$\mathrm{ad}_A(B) = [A, B]$

$\downarrow$

$\mathrm{ad}(A)(B)$

$\mathrm{ad}_A \equiv \mathrm{ad}(A)$

Kobayashi uses ad but means Ad

$\boxed{\text{Prop 5.1}}$ A^* vector field associated to $A \in \mathfrak{g}$

$a \in G$

$R_a : P \to P$ right multiplication by a
$u \mapsto u \triangleleft a$

$u \triangleleft a \equiv R_a(u)$

<u>notation:</u> $f : M \to N$ $\qquad df_u \equiv (f_*)_u$ pushforward

$dR_a|_u : T_u P \to T_{u \triangleleft a} P$

$$\boxed{dR_a|_u (A^*|_u) = (a^{-1} A a)^*|_{u \triangleleft a} \equiv \left(\mathrm{Ad}(a^{-1})(A)\right)^*|_{u \triangleleft a}}$$

Aside if $F: M \to N$ $\quad X \in T_uM$ ← derivation

$dF_u : T_uM \to T_{F(u)}N$

$dF_u(X) \in T_{F(u)}N$ → what does it look like

$h: N \to \mathbb{R}$

$$[dF_u(X)]h = X(\underbrace{h \circ F}_{M \to \mathbb{R}})$$

how to define a vector using derivations at u

$X \in T_uM$ is a linear map $C^\infty(M,\mathbb{R}) \to \mathbb{R}$

s.t. $X(fg) = f(u)X(g) + g(u)X(f)$

exercise: show that $dF_u(X)(fg) = f(F(u))(dF_u(X))(g) + g(F(u))(dF_u(X))(f)$

back to proof $\quad A^*_u f = \frac{d}{dt}\left(f(u \triangleleft e^{tA})\right)\Big|_{t=0}$ by definition

$$d(R_a)_u(A^*_u)h = A^*_u(h \circ R_a) \qquad a \in G \quad u \in P$$

$$= \frac{d}{dt}\left(h(R_a(u \triangleleft e^{tA}))\right)\Big|_{t=0}$$

$$= \frac{d}{dt}\left(h(u \triangleleft e^{tA}a)\right)\Big|_{t=0}$$

LHS

$$(a^{-1}Aa)^*_{u \triangleleft a} h = \frac{d}{dt}\left(h\left((u \triangleleft a) \triangleleft e^{ta^{-1}Aa}\right)\right)\Big|_{t=0}$$

$= a^{-1}e^{tA}a$

prove as exercise

$$= \frac{d}{dt}\left(h\left(u \triangleleft \underbrace{a \triangleleft a^{-1}}_{=e} e^{tA}a\right)\right)\Big|_{t=0}$$

$$= \frac{d}{dt}\left(h(u \triangleleft e^{tA}a)\right)\Big|_{t=0}$$

Associated bundle

suppose $\pi : P \to M$ principal G-bundle

let F be manifold on which G acts from the left

Goal: build a fibre bundle with base M and fibre F

start by looking at $P \times F$

$(gh)^{-1} = h^{-1} g^{-1}$

G acts on $P \times F$ $\quad (u, f) \in P \times F$

$(u, f) \triangleleft g = (u \triangleleft g, g^{-1} \triangleright f)$

$\Leftarrow$ check that

$((u,f) \triangleleft g) \triangleleft h = (u,f) \triangleleft (gh)$

$$E = P \times_G F := P \times F / G$$

$$= \{ [(u,f)] \mid u \in P, f \in F \} \qquad [(u,f)] = \{ (u,f) \triangleleft g \mid g \in G \}$$

projection? $\quad \pi_E : E \to M$

$[(u,f)] \mapsto \pi(u)$ $\Leftarrow$ well defined because

$\pi(u \triangleleft g) = \pi(u)$

is E a manifold? $\to$ YES

$\boxed{\text{Connections}}$

the A^* vector fields are vertical

$\to$ their integral curves are contained in the fibres

$\dot{\gamma}(t) = A^*_{\gamma(t)}$

at each point in $T_u P$ we define a subspace $V_u = \text{span}\{ A^*_u \mid A \in \mathfrak{g} \}$

what we want is to write $T_u P = V_u \oplus H_u$

Horizontal subspace

$\mathbb{R}^2 = \text{span}\{(1,0)\} \oplus \text{span}\{(0,1)\}$

$= \text{span}\{(1,0)\} \oplus \text{span}\{(1,1)\}$

in choosing H_u we want to make sure that it plays nice with group action.

$(dR_a)_u(V_u) = V_{u \triangleleft a}$ ← consequence of 5.1

→ requirement: $\boxed{(dR_a)_u(H_u) = H_{u \triangleleft a}}$

also want the mapping $u \to H_u$ to be smooth (there a bundle H) with $H_u = \pi^{-1}(\{u\})$